Stable Homotopy Theory

IAS/PCMI Summer School

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These are personal lecture notes on Professor Anna Marie Bohmann's lecture series on stable homotopy theory at the PCMI Undergraduate Summer School. These notes may include errors, but we hope they are helpful to the reader nonetheless. Feel free to inform us of any such errors – either in-person or through our emails: shawap@rose-hulman.edu or ravishs@rose-hulman.edu.

We start by assuming a background in topological spaces and the fundamental group (days 1 and 2). See [May99] or [Hat02] for more information regarding these topics.

1. Higher Homotopy groups (day 3)

1.1. **Definition.** Let $f : X \to Y$ be a map and $H : X \times I \to Y$ be a homotopy. We say f is a based map if it takes the basepoint of X to the basepoint of Y and we say H is a based homotopy if it is a based map for all $t \in I$.

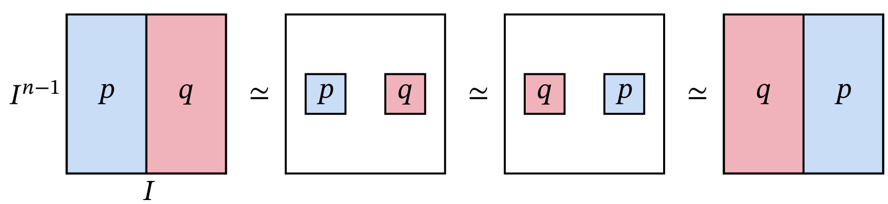
1.2. **Notation.** Let X and Y be be based spaces. Then, we define $[X, Y]$ to be homotopy classes of based maps from X to Y.

1.3. **Definition** Higher homotopy groups**.** Let X be a space and n be a nonnegative integer. The nth homotopy group of X is $\pi_n(X) \coloneqq [S^n, X]$.

Note that for the $n = 0$ case, $S^n = S^0 = \{\pm 1\}$, so $\pi_0(X)$ counts the path components of X.

Our next goal is to study the structure on $\pi_n(X)$. It turns out that for $n \neq 0$, $\pi_n(X)$ is a group and if $n > 1$, then $\pi_n(X)$ is abelian.

1.4. **Exercise.** Show the above claim! Note that if $p, q \in \pi_n(X)$, then $p * q$ is defined similarly to the group structure as on $\pi_q(X)$, so the domains of p and q can be rearranged as follows.



What else can we do with π_n? If we have a map $f : X \to Y$, then there exists a homomorphism $f_* : \pi_n(X) \to \pi_n(Y)$ given by $f_*[p] = [f \circ p]$. If $f \simeq g$, then $f_* = g_*$.

Let's do some calculations!

1.5. **Example.** Claim: $\pi_n(\mathbb{R}^m) = 0$ for all $n, m \in \mathbb{Z}^+$. This follows directly from the fact that we can explicitly construct a homotopy between $\mathrm{Id}_{\mathbb{R}^m}$ and constant map $\mathrm{Id}_{\{*\}}$ for any $* \in \mathbb{R}^m$. In particular, $H(x, t) = (1 - t)\mathrm{Id}_{\{*\}}(x) + t\mathrm{Id}_{\mathbb{R}^m}(x)$ suffices.

1.6. **Example.** Claim: $\pi_n(S^1) = 0$ for $n \geq 2$. Since $\mathbb{R}$ is a covering space of S^1 and we can take S^1 to be embedded in S^n (as well as other technical details), we can lift the projection map $p : S^n \to S$ for $n \geq 2$ to get a map $\tilde{p} : S^n \to \mathbb{R}$. This map respects hometowns and since $\pi_n(\mathbb{R}) = 0$, we know that $\pi_n(\S^1) = 0$ too.

Finally, we have that π_n preserves products:

1.7. **Theorem.** *Let X and Y be topological spaces. Then $\pi_n(X \times Y) = \pi_n(X) \times \pi_n(Y)$.*

Proof outline. Since elements of $\pi_n(X)$ and $\pi_n(Y)$ are homotopy classes of based maps from S^n to the respective space, we can essentially use the universal property of products to get that (based homotopy) maps from $S^n \to X \times Y$ correspond to maps $S^n \to X$ and $S^n \to Y$.

1.8. **Proposition.** *$\pi_i(S^n) = 0$ for $i < n$.*

Proof. We can approximate any map $S^i \to S^n$ by a smooth/cellular map to see that there is a homotopy of the map which misses a point. $S^n \setminus \{p\}$ is homeomorphic to $\mathbb{R}^n$ by stereographic projection; since $\mathbb{R}^n$ is contractible, we can contract the map to a point.

2. Constructions on based spaces (day 4)

Wedge sum: $X \vee Y \coloneqq X \sqcup Y/\{*\}$
Smash product: $X \wedge Y \coloneqq X \times Y/X \vee Y$
(Reduced) Cone: $CX \coloneqq X \wedge I$
(Reduced) Suspension: $\Sigma X \coloneqq X \wedge S^1$ like two cones glued one on top of the other
(Based) Loop Space: $\Omega X \coloneqq \mathrm{Hom}_{\mathsf{Top}_*}(S^1, X)$ with the compact-open topology
Path Space: $PX \coloneqq \mathrm{Hom}_{\mathsf{Top}_*}((I,0),(X,*))$ with the compact-open topology

2.1. **Theorem.** *Σ and Ω are adjoint functors i.e. there is a natural isomorphism*

$$\mathrm{Hom}_{\mathsf{Top}_*}(\Sigma X, Y) \cong \mathrm{Hom}_{\mathsf{Top}_*}(X, \Omega Y).$$

2.2. **Corollary.** $\pi_n(X) \cong [S^0, \Omega^n X]$.

Proof.

$$\pi_n(X) = [S^n, X] \cong [S^1 \wedge S^{n-1}, X] \cong [\Sigma S^{n-1}, X] \cong [S^{n-1}, \Omega X] \cong \cdots \cong [S^0, \Omega^n X]$$

Note that $[S^0, \Omega^n X] = \pi_0(\Omega^n X)$; though π_0 typically does not carry a group structure we do obtain a group structure for loop spaces.

2.3. **Definition.** A fibration is a map $p \colon E \to B$ such that for any Y and maps $f \colon Y \to E$, $h \colon Y \times I \to B$ for which the diagram

$$\begin{tikzcd} Y \arrow[r, "f"] \arrow[d, hook, "\iota_0"'] & E \arrow[d, "p"] \\ Y \times I \arrow[r, "h"] \arrow[ur, dashed, "\tilde{h}"] & B \end{tikzcd}$$

commutes, there exists a lift $\tilde{h} \colon Y \times I \to E$. The space E is called the total space and B is called the base space.

2.4. **Proposition.** (1) *We have a map $p_1 \colon PX \to X$ given by $p \mapsto p(1)$.*

(2) *PX is contractible by the homotopy $H \colon PX \times I \to PX$ taking $(q,t) \mapsto q|_{[0,t]}$. Thus $\pi_n(PX) = 0$ for all $n \geq 1$.*

(3) *The fibre of the basepoint under p_1 is ΩX. This gives a fibration*

$$\Omega X \hookrightarrow PX \xrightarrow{p_1} X.$$

If $E \xrightarrow{p} B$ is a fibration, we have a sequence of maps

$$\cdots \longrightarrow \Omega^2 B \longrightarrow \Omega F \longrightarrow \Omega E \xrightarrow{\Omega p} \Omega B \longrightarrow F \longrightarrow E \xrightarrow{p} B.$$

Then, we have the following.

2.5. **Lemma.** *For any based space Z, we get the exact sequence*

$$\cdots \longrightarrow [Z, \Omega E] \longrightarrow [Z, \Omega B] \longrightarrow [Z, F] \longrightarrow [Z, E] \longrightarrow [Z, B].$$

To make sense of this sequence being exact, we recall that $[Z, \Omega X] \cong [\Sigma Z, X]$ has a group structure. The last three terms are not groups but can be thought of as pointed sets. For maps into these spaces, we take the kernel to be the preimage of the class of the constant map.

2.6. **Theorem.** *For a fibration $F \to E \to B$ where B is path connected, we have a long exact sequence:*

$$\cdots \to \pi_n(F) \to \pi_n(E) \to \pi_n(B) \to \pi_{n-1}(F) \to \cdots \to \pi_1(B) \to \pi_0(F) \to \pi_0(E) \to \pi_0(B)$$

Proof. Apply Lemma 2.5 with $Z = S^0$.

2.7. **Example** Hopf bundle. Let $\mathbb{CP}^1$ denote the complex projective line, which is homeomorphic to S^2. We get the Hopf map $\eta : S^3 \to \mathbb{CP}^1$ mapping $(z_1, z_2) \mapsto [z_1, z_2]$ where we view S^3 as the unit sphere in $\mathbb{C}^2$. Composing with the homeomorphism $\mathbb{CP}^1 \xrightarrow{\sim} S^2$ gives a fibre bundle with fibre S^1. We can see that η is a fibration with fiber S^1. Note that

$$\eta^{-1}([0,1]) = \{(0, z_2) \in \mathbb{C}^2 : |(0, z_2)| = 1\} = \{(0, z_2) \in \mathbb{C}^2 : |z_2| = 1\} \simeq S^1$$

Applying the long exact sequence from Theorem 2.6, we get

$$\cdots \longrightarrow \pi_2(S^3) \longrightarrow \pi_2(S^2) \longrightarrow \pi_1(S^1) \longrightarrow \pi_1(S^3) \longrightarrow \cdots$$

Since $\pi_2(S^3) = \pi_1(S^2) = 0$ (from Proposition 1.8) and $\pi_1(S^1) = \mathbb{Z}$ we get that $\pi_2(S^2) = \mathbb{Z}$!

We were able to calculate this higher homotopy group because of a long exact sequence! Let's define another one. For a based inclusion $A \hookrightarrow X$, we can define

$$P(X; *, A) = \{\text{paths } g : I \to X : g(0) = * \text{ and } g(1) \in A\}.$$

2.8. **Definition** Relative homotopy groups. Given the above setup, define $\pi_n(X, A) = \pi_{n-1}(P(X; *, A))$.

2.9. **Example** Sanity check. If $A = *$, then this becomes $\pi_n(X, *) = \pi_{n-1}(\Omega X)$ since $P(X; *, *) = \Omega X$.

In general, we think of these as maps $I^n \to X$ such that part of ∂I^n goes to $*$ and the rest goes to A. Why do we care? This yields the fibration

$$\cdots \longrightarrow \Omega P(X; *, A) \longrightarrow \Omega A \longrightarrow \Omega X \longrightarrow P(X; *, A) \xrightarrow{p_1} A \longrightarrow X \tag{2.10}$$

which, in turn, yields

$$\cdots \longrightarrow \pi_2(X, A) \longrightarrow \pi_1(A) \longrightarrow \pi_1(X) \longrightarrow \pi_1(X, A) \longrightarrow \pi_0(A) \longrightarrow \pi_0(X)$$

by Theorem 2.6. See the exercises for applications.

Recall that if two maps $f : X \to Y$ and $g : Y \to X$ are homotopy equivalences, then f_* and g_* induce isomorphisms on π_n for all n. We can define a weaker notion of this:

2.11. **Definition.** A map $e : Y \to Z$ is an n-equivalence if for all $y \in Y$, then $e_* : \pi_q(Y, y) \to \pi(Z, e(y))$ is an isomorphism for all $q < n$ and an epimorphism on $q = n$. If e_* is an isomorphism for all q then we say e is a *weak equivalence*.

2.12. **Definition.** X is n-connected if $\pi_q(X, x) = 0$ for all $0 \leq q \leq n$

Similar definitions follow for pairs (where in the analogue of the second definition, we demand $\pi_0(A) \simeq \pi_0(X)$).

2.13. **Theorem.** *(X, A) is n-connected if and only if $A \hookrightarrow X$ is an n-equivalence.*

Proof. From Theorem 2.6, we have the long exact sequence

$$\cdots \longrightarrow \pi_{q+1}(X) \longrightarrow \pi_{q+1}(X,A) \longrightarrow \pi_q(A) \longrightarrow \pi_q(X) \longrightarrow \pi_q(X,A) \longrightarrow \pi_{q-1}(A) \longrightarrow \cdots$$

If (X,A) is n-connected then for each $q < n$ we have $\pi_q(X,A) = 0$. The exact sequence becomes

$$\cdots \longrightarrow 0 \longrightarrow \pi_q(A) \longrightarrow \pi_q(X) \longrightarrow 0 \longrightarrow \cdots$$

so the maps $\pi_q(A) \to \pi_q(X)$ are isomorphisms. For $q = n$ we have

$$\cdots \longrightarrow \pi_{n+1}(X,A) \longrightarrow \pi_n(A) \longrightarrow \pi_n(X) \longrightarrow 0 \longrightarrow \cdots$$

so the map $\pi_n(A) \to \pi_n(X)$ is a surjection. By definition, this means $A \hookrightarrow X$ is an n-equivalence.

Conversely, if $A \hookrightarrow X$ is an n-equivalence then the maps $\pi_{q+1}(X,A) \to \pi_q(A)$ and $\pi_q(X) \to \pi_q(X,A)$ are zero maps. For $q \le n$ we have zero maps in and out of $\pi_q(X,A)$ so exactness implies that $\pi_q(X,A) = 0$.

How exciting!

2.14. **Theorem** Homotopy excision**.** *Let X be a space with subsets A and B such that $X = \operatorname{Int} A \cup \operatorname{Int} B$ and $C := A \cap B \neq \varnothing$. Assume both (A,C) is $(m-1)$-connected and (B,C) is $(n-1)$-connected where $m \ge 2$ and $n \ge 1$. Then, the inclusion $(A,C) \hookrightarrow (X,B)$ induces a homomorphism $\pi_q(A,C) \to \pi_q(X,B)$, which is an isomorphism for $q < m+n-2$ and a surjection for $q = m+n-2$ (i.e., a $(m+n-2)$-equivalence).*

2.15. **Corollary.** *Let $i : A \hookrightarrow X$ be a "nice" inclusion and a $(n-1)$-equivalence. Furthermore, suppose X, A are both $(n-2)$-connected for $n \ge 2$. Then, the quotient map $(X,A) \to (X/A, *)$ is an $(2n-2)$-equivalence. Moreover, the quotient map is a $(2n-1)$-equivalence if A and X are $(n-1)$-connected.*

Let X be a based space. There is a suspension homomorphism $\Sigma : \pi_q(X) \to \pi_{q+1}(\Sigma X)$ which takes $f : S^q \to X$ to $f \wedge \mathrm{Id}_{S^1} : S^q \wedge S^1 = S^{q+1} \to X \wedge S^1 = \Sigma X$.

3. Freudenthal Suspension and Pontryagin–Thom (day 6)

3.1. **Theorem** Freudenthal Suspension Theorem. *Assume the basepoint is nice and X is $n-1$ connected for $n \geq 1$. Then $\Sigma : \pi_q(X,*) \to \pi_{q+1}(\Sigma X,*)$ is an isomorphism for $q < 2n-1$ and an epimorphism for $q = 2n-1$.*

Proof. We begin with an alternative description of the map Σ: recall that the cone of X is defined by

$$CX = X \wedge I = \frac{X \times I}{(X \times \{0\}) \cup (\{*\} \cup I)},$$

so then we can easily see that $CX/(X \times \{1\}) = \Sigma X$. Then if we consider the map $f : (I^q, \partial I^q) \to (X,*)$ in $\pi_q(X)$ and then the subsequent product $f \times \mathrm{Id}_I : I^{q+1} \to X \times I$, then can then form the quotient map $X \times I \to CX$. This yields another map $(I^{q+1}, \partial I^{q+1}, I^q) \to (CX, X, *)$. Finally taking the quotient map $\rho : CX/X \times \{1\} \to \Sigma X$ we get the map Σf. We thus have a commutative diagram

$$\begin{tikzcd} & \pi_{q+1}(CX,X) \arrow[dl, "\partial"'] \arrow[dr, "\rho_*"] & \\ \pi_q(X,*) \arrow[rr, "\Sigma"'] & & \pi_{q+1}(\Sigma X,*) \end{tikzcd}$$

Since CX is contractible, the connecting homomorphism ∂ is an isomorphism. Since $X \hookrightarrow CX$ is a nice inclusion, by Corollary 2.15 the quotient map is a $2n$-equivalence $(CX,X) \to (\Sigma X,*)$. Thus Σ is also an isomorphism. 🐘

This is amazing! What a fun theorem! We had to assume a niceness of X (i.e., $(n-1)$-connectivity), but we get quite the return on investment since the suspension isomorphism theorem yields info all the way up to $q = 2n-1$.

3.2. **Example.** $\pi_n(S^n) = \mathbb{Z}$ for all $n \geq 1$. We know that $\pi_1(S^1) = \pi_2(S^2) = \mathbb{Z}$. Since S^2 is 1-connected, we can take $n = 2$ in the Freudenthal Suspension Theorem so $\pi_2(S^2) \cong \pi_3(S^3)$. Induction on n yields the result since S^n is always $(n-1)$-connected.

3.3. **Corollary.** $\pi_k(S^k) \simeq \mathbb{Z}$ *for all* $k \in \mathbb{Z}^+$.

Proof. Example 3.2. 🐘

3.4. **Consequence** (Stability #1).

$$\pi_q(X) \longrightarrow \pi_{q+1}(\Sigma X) \longrightarrow \pi_{q+2}(\Sigma^2 X) \longrightarrow \cdots \longrightarrow \pi_{q+n}(\Sigma^n X) \longrightarrow \cdots$$

if $q < n-1$, these maps become isomorphism so we can define

$$\pi_q^{\mathrm{st}} := \operatorname{colim}_n \pi_{q+n}(\Sigma^n X) = \pi_{q+m}(\Sigma^m X)$$

for $m >> q$.

3.1. Pontryagin–Thom construction.

Goal: relate (smooth) manifolds to stable homotopy groups.

3.5. **Theorem** Whitney Embedding Theorem. *Every smooth n-manifold with or without boundary admits a proper smooth embedding into $\mathbb{R}^{2n+1}$.*

Proof. [Lee12, Theorem 6.15]. 🐘

3.6. **Notation.** For a given smooth n-manifold M, we use TM to denote the tangent bundle of M which is a smooth manifold. By Theorem 3.5, we can embed M and TM into $\mathbb{R}^{n+k}$ for $k \in \mathbb{Z}^+$ sufficiently large. Then, we define the normal bundle NM of M to be bundle that fiber-wise satisfies the following short exact sequence.

$$0 \longrightarrow T\mathbb{R}^{n+k} \longrightarrow T\mathbb{R}^{n+k}|_M \longrightarrow TN \longrightarrow 0.$$

We will take it without proof that NM is homeomorphic to some tubular neighborhood about $M \subseteq \mathbb{R}^{n+k}$.

3.7. **Definition.** A smooth n-manifold is *framed* if there exists an n-tuple of nowhere vanishing vector fields on M which form a basis at each point of the tangent space of M.

Now! We can relate this back to spheres. Take the one-point compactification of $\mathbb{R}^{n+k}$ to obtain S^{n+k} which yields an embedding of M into S^{n+k}. Furthermore, we can also squash everything outside NM to the point at ∞. This is *not* a sphere! We assume this process yields an orientable manifold, so there exists a choice of framing (almost definitionally), which then induces a map from $S^{n+k} \to S^k$ (since the normal bundle is S^k fibre-wise).

Since we embedded M into $\mathbb{R}^{n+k}$, we can trivially embed it into $\mathbb{R}^{n+k+1}$, so we have a suspension of the map $S^{n+k} \to S^k$.

4. CW-COMPLEXES (DAY 7)

4.1. **Definition** CW complexes**.** A *CW complex* is a topological space which we build up by gluing on *cells* of increasing dimension. In particular, we inductively define the topological spaces X^k through the following

$$\begin{tikzcd} S^k \arrow[r, "f"] \arrow[d, hook] & X^k \arrow[d] \\ D^{k+1} \arrow[r] & X^k \cup_f D^{k+1} =: X^{k+1} \end{tikzcd}.$$

Note that we start with a discrete space X^0 (and there are additional regularity assumptions).

4.2. **Examples.** We can construct the sphere as a CW complex in two different ways. We can construct the sphere by taking two 0-cells, attaching two 1-cells to make a circle, and then attaching two 2-cells to complete the construction. Similarly, we can construct the sphere by taking a 0-cell and attaching a 2-cell to it.

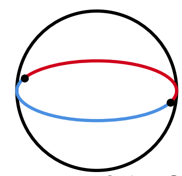


Diagram of the first construction.

4.3. **Definition.** The *dimension* of a CW complex is the dimension of the highest degree n-cell.

4.4. **Theorem** (Whitehead)**.** *If X is a CW complex and $e: Y \to Z$ is an n-equivalence, then the induced (unbased) map $[X,Y] \to [X,Z]$ is a bijection if $\dim X < n$ and a surjection if $\dim X = n$ (where n can equal infinity).*

Proof outline. We perform induction on number of cells and consider only surjectivity. Given a map $e: Y \to Z$, if we also have a map $X^k \to Z$ that we can lift to Y, we just need to lift to each $(k+1)$-cell. To understand how the $(k+1)$-cells behave, note that they are given by maps $S^k \to X^k$ and thus need to understand $S^k \to Z$ versus $S^k \to Y$, which we are given by assumption.

4.5. **Corollary** (Also Whitehead's theorem)**.** *An n-equivalence between CW complexes of dimension $< n$ is a homotopy equivalence. A weak equivalence between CW complexes is also a homotopy equivalence.*

Proof. Weak equivalence implies homotopy equivalence: if $e: Y \to Z$ is a weak equivalence between CW complexes, then its induced map $[Z,Y] \to [Z,Z]$ is a bijection, so there exists a map $f: Z \to Y$ such that $e \circ f = \mathrm{Id}_Z$. Then $e \circ f \circ e = \mathrm{Id}_Z \circ e = e$. Since $[Y,Y] \to [Y,Z]$ is also a bijection (picking a different induced map), we know $e \circ \mathrm{Id}_Y = e$, so $e \circ (f \circ e) \simeq e$. Thus, we also have $\mathrm{Id}_Y = f \circ e$.

4.6. **Theorem** Two approximation theorems**.**

(1) *If $f: X \to Y$ is a map between CW complexes, then f is homotopic to a cellular map (i.e., a map that sends n-skeletons to n-skeletons). Additionally, any homotopy $H: X \times I \to Y$ between cellular maps is equivalent to a cellular homotopy (i.e., H is a cellular map).*

(2) *For any space X, there exists a CW complex ΓX and a weak equivalence $\gamma: \Gamma X \to X$. If X is n-connected we can take ΓX to have one 0-cell and no k-cells for $1 \le k \le n$.*

Note that relative versions (like (X,A)) of these theorems also hold.

These are *extremely* strong. This (not trivially) implies that we can define algebraic invariants of general topological spaces by defining it on CW complexes.

4.7. **Theorem** (Cellular Homology exists!!!). *Fix an abelian group G. For each $q \in \mathbb{Z}$, there exists*

(1) *an abelian group $H_q(X, A; G)$ corresponding to a pair of CW complexes (X, A);*

(2) *a group homomorphism $f_* : H_q(X, A; G) \to H_q(Y, B; G)$ for each cellular map $f : (X, A) \to (Y, B)$ such that $f_* = g_*$ if $f \simeq g$ and $(\mathrm{Id}_{(X,A)})_* = \mathrm{Id}_{H_q(X,A;G)}$, and $(f \circ g)_* = g_* \circ f_*$;*

(3) *natural boundary maps $\partial : H_q(X, A; G) \to H_{q-1}(A, \varnothing; G) =: H_{q-1}(A; G)$ which satisfy the Eilenberg–Steenrod axioms:*

Dimension: *$H_q(*; G)$ equals G if $q = 0$ and 0 otherwise;*

Exactness: *the following sequence is exact*

$$\cdots \xrightarrow{\partial} H_q(A; G) \longrightarrow H_q(X; G) \longrightarrow H_q(X, A; G) \xrightarrow{\partial} H_{q-1}(A; G) \longrightarrow \cdots;$$

Excision: *If $X = A \cup B$ where A and B are subcomplexes, then the map $(A, A \cap B) \to (X, B)$ induces isomorphism $H_q(A, A \cap B; G) \to H_q(X, B; G)$;*

Additivity: *if (X, A) is the disjoint union of (X_i, A_i), then*

$$\bigoplus_i H_q(X_i, A_i; G) \simeq H_q(X, A; G).$$

4.8. **Corollary.** *For each nonnegative integer q, we get a functor H_q from the homotopy category $\mathsf{CW\text{-}Pairs}_{\mathsf{Htpy}}$ of pairs of CW complexes to Ab. In the case where the subcomplex is empty, we have functors $H_q(-; G)$ from the homotopy category of CW complexes to Ab.*

5. Cellular Homology (day 8)

Today, we will explicitly construct the cellular homology groups $H_q(X, A; G)$ where $G = \mathbb{Z}$. We will omit the $\mathbb{Z}$-dependence. We will also begin with the case $A = \varnothing$, and we will also omit the A-dependence in this case, writing $H_q(X)$ for $H_q(X, \varnothing; \mathbb{Z})$.

5.1. **Observation.** Let X^n denote the n-skeleton of X. Then for any n, we have

$$X^n/X^{n-1} \cong \bigvee_{n \text{ cells of } X} S^n.$$

Let I denote some indexing set. From the exercises, we know π_n splits over wedge sums of S^n when $n \geq 2$:

$$\pi_n\Big(\bigvee_{i \in I} S^n\Big) = \bigoplus_{i \in I} \mathbb{Z}.$$

For $n = 0$, π_0 is not necessarily a group and for $n = 1$, π_1 is not necessarily abelian, so we make the following definition before we discuss homology

$$\tilde{H}'_n(X) := \begin{cases} \bigoplus_{x \in X \setminus \{*\}} x\mathbb{Z} & \text{if } n = 0; \\ \pi_1(X)^{\mathrm{ab}} = \pi_1(X)/[\pi_1(X), \pi_1(X)] & \text{if } n = 1; \\ \pi_n(X) & \text{if } n \geq 2. \end{cases}$$

We wish to check that $\tilde{H}'_n(X)$ plays nicely with common operations:

Suspension: Let X be based and $(n-1)$-connected. Then we can define a suspension map $\Sigma\colon \tilde{H}'_n(X) \to \tilde{H}'_n(\Sigma X)$ by the following. If $n \geq 1$, then $\pi_n(X) \ni [f] \mapsto [\Sigma f] \in \pi_{n+1}(X)$ and for $x \in \tilde{H}'_0(X)$ we set $\Sigma[x] := [f_*^{-1} - f_x]$.

5.2. **Lemma.** *For an indexing set I, $\Sigma\colon \tilde{H}'_n(\bigvee_{i \in I} S^n) \to \tilde{H}'_{n+1}(\Sigma(\bigvee_{i \in I} S^n))$ is an isomorphism.*

Proof. Since S^n is compact, any map $S^n \to \bigvee_I S^n$ has compact image; in particular the image must be contained in the wedge of finitely many spheres. Thus it suffices to show this when I is finite. So, suppose I is finite, which implies that $\tilde{H}'_n(\bigvee_{i \in I} S^n) = \oplus_{i \in I} \mathbb{Z}$ and $\tilde{H}'_{n+1}(\Sigma(\bigvee_{i \in I} S^n)) = \oplus_{i \in I} \mathbb{Z}$, so Σ is just an isomorphism by Theorem 3.1.

Returning to our previous discussion: to understand X (a CW complex) we need to understand how X^n/X^{n-1} and X^{n-1}/X^{n-2} relate (we make the convention that $X^0/X^{-1} = X^0 \cup \{\mathrm{pt}\}$). To do so, let $i\colon X^{n-1} \hookrightarrow X^n$ denote the inclusion. Next, we define a generalization of the cone on X, called the cone on i, denoted by Ci which glues CX^{n-1} to X^n via i. Then, we have a homotopy equivalence between Ci and X^n/X^{n-1}, which we will denote by ψ. Let ψ^{-1} be a map such that $\psi \circ \psi^{-1}$ and $\psi^{-1} \circ \psi$ are homotopic to the respective identity maps. Additionally, we define the map $\pi\colon Ci \to \Sigma X^{n-1}$ by taking the quotient by the copy of X^n in Ci.

Next, we define a composite map

$$\partial_n\colon X^n/X^{n-1} \xrightarrow{\psi^{-1}} Ci \xrightarrow{\pi} \Sigma X^{n-1} \xrightarrow{\Sigma\rho} \Sigma(X^{n-1}/X^{n-2})$$

Note that $\Sigma(X^{n-1}/X^{n-2})$ is a suspension of $(n-1)$-cells.

5.3. **Definition.** We define $C_q(X) = \tilde{H}'_q(X^q/X^{q-1}) = \oplus_{q\text{-cells of } X} \mathbb{Z}$, which we call the q-chains of X. We commonly think of this as the free abelian group on the q-cells of X, as we have on the right. Next, we can define the map $d_q : C_q(X) \to C_{q-1}(X)$ by the composition

$$\tilde{H}'_q(X^q/X^{q-1}) \xrightarrow{(\partial_q)_*} \tilde{H}'_q(\Sigma(X^{q-1}/X^{q-2})) \xrightarrow{\Sigma^{-1}} \tilde{H}'_{q-1}(X^{q-1}/X^{q-2}).$$

Secretly, this is a chain complex: a sequence of abelian groups C_i and maps $d_i : C_i \to C_{i-1}$ such that $d_{i-1} \circ d_i = 0$ for all i. In particular we have $\operatorname{im} d_i \subset \ker d_{i-1}$, so any exact sequence is a chain complex.

5.4. **Definition.** Given a chain complex $(C_\bullet, d_\bullet)$,

$$H_i(C_\bullet) := \ker d_i / \operatorname{im} d_{i+1}$$

is the homology of $(C_\bullet, d_\bullet)$.

Back to our cellular chains $C_q(X)$! Why do we have $d_{q-1} \circ d_q = 0$? This is the composite of the following sequence of maps:

$$X^q \cup_i CX^{q-1} \xrightarrow{\pi} \Sigma X^{q-1} \longrightarrow \Sigma(X^{q-1} \cup CX^{q-2}) \longrightarrow \Sigma(\Sigma X^{q-2})$$

which can be pictorially represented as

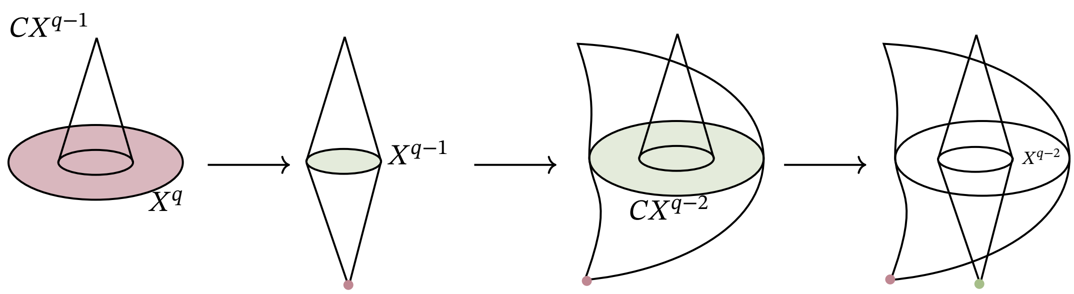


FIGURE 1. Pictorial representation of the composition of maps.

We can now define our homology groups:

5.5. **Definition.**

$$H_q(X) := H_q(C_\bullet(X), d_\bullet) = \ker d_q / \operatorname{im} d_{q+1}.$$

5.6. **Example** Homology of S^2. We can construct S^2 as a CW complex by gluing a single 0-cell to a 2-cell. This means $C_0(S^2) = \mathbb{Z}$, $C_2(S^2) = \mathbb{Z}$, and $C_q(S^2) = 0$ for all $q \neq 0, 2$. We thus have the chain complex

$$\cdots \longrightarrow 0 \longrightarrow \mathbb{Z} \longrightarrow 0 \longrightarrow \mathbb{Z} \longrightarrow 0.$$

This immediately yields

$$H_q(S^2) = \begin{cases} \mathbb{Z} & \text{if } q = 0, 2; \\ 0 & \text{otherwise.} \end{cases}$$

5.7. **Example** Homology of S^1. We have $C_1(S^1) = C_0(S^1) = \mathbb{Z}$ and $C_q(S^1) = 0$ for $q \geq 2$. This gives us the complex

$$\cdots \longrightarrow 0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow 0; \ .$$

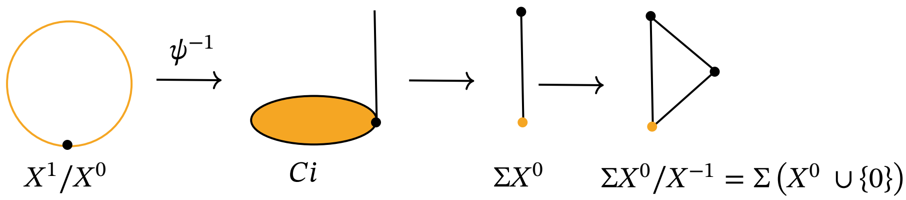


FIGURE 2. The boundary map $d_1 : C_1(S^1) \to C_0(S^1)$

Per the figure above, the map $\mathbb{Z} \to \mathbb{Z}$ is the zero map so our homology groups are

$$H_q(S^1) = \begin{cases} \mathbb{Z} & \text{if } q = 0, 1; \\ 0 & \text{otherwise} \end{cases}$$

We claimed in Theorem 4.7 that the homology groups would satisfy certain axioms; it remains to show that the groups we have defined here are the ones claimed to exist in the theorem. First we define relative homology: given an inclusion of CW complexes $A \hookrightarrow X$, we define

$$C_\bullet(X, A) \coloneqq C_\bullet(X)/C_\bullet(A);$$

then the relative homology groups $H_q(X, A)$ are the homology groups of the chain complex $C_\bullet(X, A)$. We will now discuss why this satisfies the aforementioned axioms.

Cellular map (functoriality): Given a cellular map $f : X \to Y$, we have an induced map $X^n/X^{n-1} \to Y^n/Y^{n-1}$. This yields a map of chain complexes $C_\bullet(X) \xrightarrow{f_\bullet} C_\bullet(Y)$. This "map" is a commutative diagram

$$\begin{tikzcd} \cdots \arrow[r, "d_{q+2}"] & C_{q+1}(X) \arrow[r, "d_{q+1}"] \arrow[d, "f_{q+1}"] & C_q(X) \arrow[r, "d_q"] \arrow[d, "f_q"] & C_{q-1}(X) \arrow[r, "d_{q-1}"] \arrow[d, "f_{q-1}"] & \cdots \\ \cdots \arrow[r, "d_{q+2}"'] & C_{q+1}(Y) \arrow[r, "d_{q+1}"'] & C_q(Y) \arrow[r, "d_q"'] & C_{q-1}(Y) \arrow[r, "d_{q-1}"'] & \cdots \end{tikzcd}$$

which yields a well-defined map on homology (see exercises).

Dimension: If X is a single point we have $C_0(X) = \mathbb{Z}$ and $C_q(X) = 0$ for $q \geq 1$. This yields $H_0(X) = \mathbb{Z}$ and $H_q(X) = 0$ for $q \geq 1$;

Additivity: If $X = \coprod_{i \in I} X_i$ then $C_\bullet(X) = \bigoplus_{i \in I} C_\bullet(X_i)$, which yields $H_q(X) = \bigoplus_{i \in I} H_q(X_i)$;

Excision: Recall $C_\bullet(X)/C_\bullet(A) = C_\bullet(X, A)$ and observe that if $X = A \cup B$ then $C_\bullet(X)/C_\bullet(A) = C_\bullet(B)/C_\bullet(A \cap B)$;

Exactness: By definition we have the short exact sequence of chain complexes

$$0 \longrightarrow C_\bullet(A) \longrightarrow C_\bullet(X) \longrightarrow C_\bullet(X, A) \longrightarrow 0$$

which is a commutative diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & \\
\cdots \arrow[r, "d_{q+1}"] & C_q(A) \arrow[r, "d_q"] \arrow[d] & C_{q-1}(A) \arrow[r, "d_{q-1}"] \arrow[d] & \cdots \\
\cdots \arrow[r, "d_{q+1}"] & C_q(X) \arrow[r, "d_q"] \arrow[d] & C_{q-1}(X) \arrow[r, "d_{q-1}"] \arrow[d] & \cdots \\
\cdots \arrow[r, "d_{q+1}"'] & C_q(X,A) \arrow[r, "d_q"'] \arrow[d] & C_{q-1}(X,A) \arrow[r, "d_{q-1}"'] \arrow[d] & \cdots \\
 & 0 & 0 &
\end{tikzcd}$$

where each column is a short exact sequence of abelian groups. It is a standard fact that such a short exact sequence of chain complexes induces a long exact sequence on homology groups (zigzag lemma):

$$\cdots \xrightarrow{\partial} H_q(A) \longrightarrow H_q(X) \longrightarrow H_q(X,A) \xrightarrow{\partial} H_{q-1}(A) \longrightarrow \cdots;$$

We can realize this topologically as well: a map $X/A \xrightarrow{\psi^{-1}} Ci \to \Sigma A$ corresponds to a homomorphism $H_q(X,A) \to H_{q-1}(A)$.

6. Connections between homology and homotopy (Day 9)

6.1. **Definition** Reduced Homology**.** Let X be a based space where the basepoint $*$ is a nice inclusion. Define the reduced homology

$$\tilde{H}_q(X) \coloneqq H_q(X, *).$$

An immediate consequence is that if $A \hookrightarrow X$ then $\tilde{H}_q(X/A) \cong H_q(X, A)$.

6.2. **Theorem.** *Let X be a based topological space. Then, there is a natural isomorphism*

$$\Sigma : \tilde{H}_q(X) \xrightarrow{\sim} \tilde{H}_{q+1}(\Sigma X)$$

Note that this contrasts the π_q analogue Theorem 3.1, which has assumptions about connectivity.

Proof. Use the LES

$$\begin{tikzcd}
H_{q+1}(X,*) \arrow[r] \arrow[d, phantom, "\wr"] & H_{q+1}(CX,X) \arrow[r, "\partial"] & H_q(X,*) \arrow[r] \arrow[d, phantom, "\wr"] & H_q(CX,*) \arrow[d, phantom, "\wr"] \\
\tilde{H}_{q+1}(CX) \arrow[r] & \tilde{H}(\Sigma X) \arrow[u, "\simeq \text{ via cones}"] \arrow[r, "\sim"] & \tilde{H}_q(X) \arrow[r] & \tilde{H}_q(CX)
\end{tikzcd}$$

By the dimension axiom for reduced homology gives us a generator i_0 of $\tilde{H}_0(S^0) = \mathbb{Z}$. Using the suspension isomorphisms we inductively choose generators i_n of $\tilde{H}_n(S^n)$ by setting $i_{n+1} \coloneqq \Sigma i_n$.

6.3. **Definition** Hurewicz homomorphism**.** For a based space X define $h : \pi_n(X) \to \tilde{H}_n(X)$ by $[f] \mapsto f_*(i_n)$ where i_n is the generator of $\tilde{H}_n(S^n) = \mathbb{Z}$ chosen above and $f_* : \tilde{H}_n(S^n) \to \tilde{H}_n(X)$ is the induced map.

Proof. As defined above, the Hurewicz homomorphism is just a function; we claim it is a homomorphism. Let $f, g : S^n \to X$ so that $[f+g] \in \pi_n(X)$ is represented by the composite map

$$S^n \longrightarrow S^n \vee S^n \xrightarrow{f \vee g} X \vee X \xrightarrow{\text{fold}} X.$$

So then we can check that the Hurewicz homomorphism is indeed a homomorphism through the following composition of maps.

$$\begin{tikzcd}
 & \tilde{H}_n(S^n) \oplus \tilde{H}_n(S^n) \arrow[d, phantom, "\parallel"] & \tilde{H}_n(X) \oplus \tilde{H}_n(X) \arrow[d, phantom, "\parallel"] & \\
\tilde{H}_n(S^n) \arrow[r] \arrow[d, phantom, "\Psi"] & \tilde{H}_n(S^n \vee S^n) \arrow[r] & \tilde{H}_n(X \vee X) \arrow[r] & \tilde{H}_n(X) \\
i_n \arrow[r, mapsto] & (i_n, i_n) \arrow[r] & (f_* i_n, g_* i_n) \arrow[r, mapsto, "+"] & f_* i_n + g_* i_n = h(f) + h(g)
\end{tikzcd}$$

$$\tilde{H}_n(S^n) \longrightarrow \tilde{H}_n(S^n \vee S^n) \longrightarrow \tilde{H}_n(X \vee X) \longrightarrow \tilde{H}_n(X)$$

The Hurewicz homomorphism is a natural transformation between the functors π_n and $\tilde{H}_n : \mathsf{Top}_* \to \mathsf{Grp}$. Moreover, the following diagram commutes for all $n \geq 0$:

$$\begin{tikzcd}
\pi_n(X) \arrow[r, "h_n"] \arrow[d, "\Sigma"'] & \tilde{H}_n(X) \arrow[d, "\Sigma"] \\
\pi_{n+1}(\Sigma X) \arrow[r, "h_{n+1}"'] & \tilde{H}_{n+1}(\Sigma X)
\end{tikzcd} \tag{6.4}$$

6.5. **Lemma.** *If $X = \bigvee_{i\in I} S^n$ then $h\colon \pi_n(X) \to \tilde{H}_n(X)$ is abelianization for $n = 1$ and an isomorphism for $n > 1$.*

Proof. This reduces to the case where $I = \{*\}$ since π_n and $\tilde{H}_n$ split over wedges of n-spheres, in which case, the conclusion follows.

6.6. **Theorem** (Hurewicz)**.** *If X is $(n-1)$-connected, then $h\colon \pi_n(X) \to \tilde{H}_n(X)$ is abelianization when $n = 1$ and an isomorphism when $n > 1$.*

Proof. Using cellular approximation (Theorem 4.6), we can consider the case when $X = X^{n+1}$ has no cells of dimension $1 \le k < n$. Then X is constructed by attaching copies of D^{n+1} to X^n (which must be a wedge of spheres), so we can reduce to the previous lemma.

6.7. **Consequence.** The first nonzero degree of $\pi_q(X)$ and $\tilde{H}_q(X)$ is the same, and h is as close as possible to an isomorphism in this degree. In particular, we always have

$$\tilde{H}_1(X) = \pi_1(X)^{\mathrm{ab}}.$$

Considering Equation (6.4) again, note that even if Theorem 6.6 implies that the two arrows connected to $\tilde{H}_n(X)$ may be isomorphisms, the other two arrows (connected to $\pi_{n+1}(\Sigma X)$) may not be isomorphisms.

6.1. **Eilenberg–MacLane spaces.** So far, we have been taking classical spaces and attempting to find their homotopy groups. Now, we plan to construct spaces so that we know exactly what their homotopy groups are.

6.8. **Definition.** We say that a topological space X is a $K(G,n)$ space (for $n \in \mathbb{Z}^+$) if

$$\pi_k(X) = \begin{cases} G & \text{if } k = n; \\ 0 & \text{if } k \neq n. \end{cases}$$

Note that by this definition, $K(G,n)$ is only defined up to weak equivalence. For $n = 1$, we can construct $K(G,1)$ for G any group, but for $n > 1$, G is necessarily abelian.

6.9. **Sketch** Construction of $K(G,1)$**.** Fix a group G; we need to build a contractible space EG on which G acts freely. To begin with, consider G as a set; then G acts freely on this set by left multiplication. However this is not contractible, so we connect things. This makes it contractible but the action is not free anymore. We missed some of the details, so we recommend the reader to [May99, Chapter 16, Section 5] to see how this is solved. It defines two important spaces, which will be used in the remainder of the proof. The first, EG, is called the total space, and $BG = EG/G$ is the classifying space of G.

This yields a fibration $G \to EG \to BG$, so using the LES on homotopy Theorem 2.6 we obtain

$$\cdots \longrightarrow \pi_1(G) \longrightarrow \pi_1(E) \longrightarrow \pi_1(BG) \longrightarrow \pi_0(G) \longrightarrow \pi_0(EG) \longrightarrow \pi_0(BG).$$

Since EG is contractible (by the omitted construction), $\pi_n(EG) = 0$ for all n. This gives isomorphisms $\pi_n(BG) \xrightarrow{\sim} \pi_{n-1}(G)$, so $\pi_1(BG) = \pi_0(G) = G$ and $\pi_k(BG) = 0$ for $k > 1$. This means BG is a $K(G,1)$.

Fun facts:

- This construction is functorial in G.
- If $\alpha\colon G \to G'$ is a group homomorphism, then we get a map $EG \to EG'$. Passing to quotients, that yields a map $f_\alpha\colon BG \to BG'$, which again, induces a map $f_{\alpha*}$ on π_1.

Conversely, a map $f : BG \to BG'$ induces a group homomorphism $f_* : \pi_1(BG) \to \pi_1(BG')$ so $[BG, BG']$ is in bijection with $\mathrm{Hom}_{\mathrm{Grp}}(G, G')$.

- If G is abelian, then multiplication $m : G \times G \to G$ is a group homomorphism, which allows us to define a group structure on BG:

  $$BG \times BG \simeq B(G \times G) \xrightarrow{f_m} BG$$

  Then, we can also iterate the application of B to get $B(BG)$, $B(B(BG))$, etc. Then

  $$\pi_q(B^n G) = \pi_{q-n+1}(BG) = \begin{cases} G & \text{if } q = n; \\ 0 & \text{if } q \neq n. \end{cases}$$

  which can be shown through the LES on homotopy Theorem 2.6 again.

7. Eilenberg–MacLane spaces and cohomology (wahoo) (day 10)

Recall, an Eilenberg–MacLane space $K(G,n)$ is a topological space where $\pi_q(K(G,n)) = G$ if $q = n$ and 0 otherwise.

7.1. **Definition** Reduced cohomology**.** Let X be a topological space; the reduced cohomology of X with coefficients in G is

$$\tilde{H}^q(X;G) = [X, K(G,q)].$$

We must verify that this definition is indeed that of a cohomology theory.

Functoriality: Given $f : X \to Y$, there is an induced map $f^* : [Y, K(G,q)] \to [X, K(G,q)]$ given by precomposition with f. By the definition of precomposition, this implies functoriality. Note that the functoriality is *contravariant*—the direction of the map reverses;

Abelianness: Recall from yesterday, we have a muallows us to define the map

$$\begin{array}{ccccccc} [X,K(G,n)] & \times & [X,K(G,n)] & \longrightarrow & [X,K(G,n)\times K(G,n)] & \xrightarrow{m_*} & [X,K(G,n)] \\ \Cup & & \Cup & & \Cup & & \\ f & & g & & (x \mapsto (f(x), g(x))) & & \end{array}$$

So this defines a product on $[X, K(G,n)]$. The identity element is given by the constant map to the basepoint in $K(G,n)$. Then, the fact that $\pi_0(K(G,n))$ is a group allows us to define inverses. Thus, $\tilde{H}^n(X;G)$ is abelian for all n.

Dimension: The dimension axiom requires that

$$\tilde{H}^q(S^0;G) = \begin{cases} G & \text{if } q = 0; \\ 0 & \text{if } q \neq 0. \end{cases}$$

To see that this holds, note that it follows from

$$\tilde{H}^q(S^0;G) = [S^0, K(G,q)] = \pi_0(K(G,q))$$

and the definition of $K(G,q)$.

Exactness: If $A \hookrightarrow X$ then we require

$$\tilde{H}^q(X/A;G) \longrightarrow \tilde{H}^q(X;G) \longrightarrow \tilde{H}^q(A;G)$$

be exact. By functoriality we have natural maps

$$[X/A, K(G,q)] \to [X, K(G,q)] \to [A, K(G,q)];$$

and exactness follows from the fact that the induced maps are the respective precompositions.

Additivity: If $X = \bigvee_{i\in I} X_i$ then we require

$$\tilde{H}^q(X;G) = \prod_{i\in I} \tilde{H}^q(X_i;G).$$

This follows from

$$[X, K(G,q)] = \left[\bigvee_{i\in I} X_i, K(G,q)\right] \cong \prod_{i\in I}[X_i, K(G,q)].$$

and basic properties of how homotopy classes of maps.

Suspension: We require
$$\tilde{H}^q(X;G) \cong \tilde{H}^{q+1}(\Sigma X;G). \tag{7.2}$$

This follows from
$$\tilde{H}^{q+1}(\Sigma X;G) = [\Sigma X, K(G,q+1)] = [X, \Omega K(G,q+1)]$$
where the second isomorphism is the adjunction between suspension and loop spaces as stated in Theorem 2.1. We need to show that $\Omega K(G,q+1) = K(G,q)$ and this follows from the long exact sequence of the loop/path fibration as in Equation (2.10)

$$\cdots \to \pi_{q+1}(PK(G,q+1)) \to \pi_{q+1}(K(G,q+1)) \to \pi_q(\Omega K(G,q+1) \to \pi_q(PK(G,q+1)) \longrightarrow \pi_q(K(G,q+1)) \longrightarrow \cdots$$

Since path spaces are contractible we have $\pi_n(PK(G,q+1)) = 0$ for all n. This yields isomorphisms $\pi_n(K(G,q+1)) \cong \pi_{n-1}(\Omega K(G,q+1))$ so we can see that $\Omega K(G,q+1)$ satisfies the definition of $K(G,q)$:
$$\pi_n(\Omega K(G,q+1)) = \begin{cases} G & \text{if } n = q; \\ 0 & \text{else.} \end{cases}$$

Note that this axiom somewhat replaces excision, which can be shown with the additional help of exactness.

Thus, $\tilde{H}^n(X;G)$ is indeed a cohomology theory. Note that exactness and additivity did not use any of the properties of $K(G,n)$, and to show that $\tilde{H}^n(X;G)$ is an abelian group, we just needed multiplication up to homotopy on each $K(G,n)$. Finally, to show suspension, we really just needed $K(G,n) \simeq \Omega K(G,n+1)$.

7.3. **Theorem** Important Implicit Theorem. *A (co)homology theory on pairs of spaces/CW complexes satisfying the axioms we specified is equivalent to a reduced (co)homology theory on nice based spaces/CW complexes satisfying the corresponding axioms. Note that to go from CW complexes to all topological spaces, we need to add the axiom that weak equivalences induce isomorphisms on (co)homology.*

We now use this theorem explicitly:

7.4. **Consequence.** If there exists another construction of a reduced cohomology theory $\tilde{E}^\bullet$ with $\tilde{E}^q(S^0) = G$ if $q = 0$ and 0 otherwise, then $\tilde{E}^\bullet$ is isomorphic to $\tilde{H}^\bullet(-;G)$.

7.5. **Definition.** For a CW complex X we take the reduced chain complex $\tilde{C}_\bullet(X)$ and dualize by applying the functor $\mathrm{Hom}_{\mathbb{Z}}(-,G)$. This gives a cochain complex, the homology of which we take as the definition of a cohomology theory on CW complexes.

We can see that the dimension axiom holds due to the following.

$$\begin{tikzcd}
\cdots & \mathrm{Hom}_{\mathbb{Z}}(\tilde{C}_2(S^0),G) \arrow[l] \arrow[d, equal] & \mathrm{Hom}_{\mathbb{Z}}(\tilde{C}_1(S^0),G) \arrow[l] \arrow[d, equal] & \mathrm{Hom}_{\mathbb{Z}}(\tilde{C}_0(S^0),G) \arrow[l] \arrow[d, equal] & 0 \arrow[l] \\
\cdots & 0 \arrow[l] & 0 \arrow[l] & G \arrow[l] & 0 \arrow[l]
\end{tikzcd}$$

As in the previous definition, we call this the cohomology, but it's closely related to (and sometimes is) the homology of the dual. We can make this relation due to the universal coefficient theorem (as in [Hat02, May99]).

Why do we care about cohomology? We can define a ring structure on cohomology!

7.6. **Claim.** *If R is a ring, we get a product map*

$$K(R,p) \times K(R,q) \to K(R,p+q).$$

Then, we get a product on cohomology by the composite map

$$[X, K(R,p)] \times [X, K(R,q)] \longrightarrow [X, K(R,p) \wedge K(R,q)] \longrightarrow [X \wedge X, K(R,p+q)] \xrightarrow{\Delta^*} [X, K(R,p+q)]$$

where $\Delta \colon X \to X \wedge X$ is given by $\Delta(x) = (x,x)$, so the induced map is from the cohomology on $X \wedge X$ to the cohomology on X.

8. Generalized cohomology theories (day 11)

Recall, we defined reduced cohomology $\tilde{H}^q(X;G) = [X, K(G,q)]$ earlier this week, which was successfully a cohomology theory due to the nice properties of $K(G,q)$. Since $\tilde{H}^q(-;G)$ is given by maps into a space, we immediately got homotopy invariance of induced maps, the exact sequence $A \to X \to X/A$, and additivity. In addition, we got the suspension axiom from the homotopy equivalence between $K(G,n)$ and $\Omega K(G,n+1)$, but the only special property of $K(G,n+1)$ that we used is what implied the dimension axiom. Thus, we define a generalized cohomology theory by generalizing the homotopy equivalence between $K(G,n)$ and $\Omega K(G,n+1)$ while still defining it as "maps into a space":

8.1. **Definition.** Suppose $\{E_n\}$ is a sequence of spaces such that $E_n \cong \Omega E_{n+1}$; we can define a generalized cohomology theory $\tilde{E}^q(-)$ by $E^q(X) := [X, E_q]$ and $f^* : \tilde{E}^q(Y) \to \tilde{E}^q(X)$ by $f^* : [Y, E_q] \to [X, E_q]$ for any $f : X \to Y$.

This satisfies the following properties:

Homotopy Invariance: If $f \simeq g$ then $f^* = g^*$.
Exact Sequence: $\tilde{E}^q(X/A) \to \tilde{E}^q(X) \to \tilde{E}^q(A)$ for any CW pair (X, A).
Additivity: $\tilde{E}^q(\bigvee_{i\in i} X_i) \cong \prod_{i\in I} \tilde{E}^q(X_i)$.

Proof. These three properties immediately follow from the definition by homotopy classes of maps we can see that this is homotopy invariant.

Suspension: $\tilde{E}^q(X) \cong \tilde{E}^{q+1}(\Sigma X)$ because $[\Sigma X, E_{q+1}] \cong [X, \Omega E_{q+1}] \cong [X, E_q]$.
Dimension: Nuh uh! (We do not require a dimension axiom)

8.2. **Definition** Spectrum. A sequence of spaces $\{E_n\}$ with homotopy equivalences $E_n \to \Omega E_{n+1}$ is called a spectrum.

We argue that every spectrum "represents" a generalized cohomology theory.

8.3. **Example** Complex K-theory. Let $U(n)$ denote the n-dimensional unitary group i.e. $n \times n$ complex matrices A such that $A^{-1} = A^*$. We have a map

$$U(n) \to U(n+1) : [A] \mapsto \begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix}$$

Taking the colimit along these maps we obtain the unitary group U. Let BU denote the classifying space of U. This sits in the fibration $E \to EU \to BU$, which induces the LES as in Equation (2.10) and implies $\Omega BU \simeq U$.

8.4. **Theorem.** $\Omega U \simeq BU \times \mathbb{Z}$.

Proof. Too complicated for this class. You can explicitly construct the map, though.

8.5. **Definition** Complex K-theory. Defined $\{K_n\}$ with $K_{2i} = BU \times \mathbb{Z}$ and $K_{2i+1} = U$ where we have $\Omega BU \times \mathbb{Z} \simeq \Omega BU \xrightarrow{\sim} U$. This spectrum represents a cohomology theory called complex K-theory: $\tilde{K}^n(X) = [X, K_n]$ and $\tilde{K}^0(X) = [X_+, BU \times \mathbb{Z}]$. Note that $\tilde{K}^0(X)$ is canonically isomorphic to the set of isomorphism classes of complex vector bundles over X.

This yields valuable algebro-topological information: we can use complex K-theory to prove that if we have a multiplication map $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ with an identity and no zero divisors, then $n = 1, 2, 4, 8$ which correspond to $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$. Another thing we can prove is that if S^n has a trivial tangent bundle, then $n = 0, 1, 3, 7$.

8.6. **Theorem** Brown Representability. *If $\{\tilde{E}^n(-)\}$ is a cohomology theory (on CW complexes) satisfying*

Homotopy invariance, exactness, addditivity, and suspension,

then there exists a spectrum $\{E_n\}$ representing $\{\tilde{E}^n(-)\}$ i.e. $\tilde{E}^n(X) = [X, E_n]$.

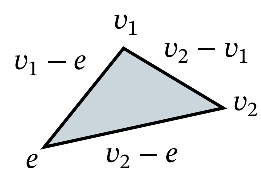
Proof outline. It is enough to show that if $\tilde{E}^q(-)$ satisfies the first three axioms then $\tilde{E}^q(-)$ is represented by $[-, E_q]$ for a space E_q. For this, we build a CW complex with "the right structure" by putting in n-cells for each element in $\tilde{E}^q(S^n)$ for all n and then fixing the higher homotopy groups. The fact that $\{E_q\}$ is a spectrum will then be enforced by the suspension axiom.

As in the case of the $K(G,n)$'s, we have a "ring-like structure" on $\{E_n\}$ which induces multiplication on $\tilde{E}^\bullet(-)$.

8.7. **Example.** For a ring R, we have structure on $K(R,n)$ as discussed in Claim 7.6:

$$K(R,p) \wedge K(R,q) \to K(R, p+q)$$

Given our construction of the $K(R,n)$'s through BR, we can construct this product for $p = 0$ and $q = 1$ (which is a map $R \times BR \to BR$). To get BR, we need an R-action on ER, which we can take to be addition and translation



Thus, the multiplication action descends to an action on BR: $R \times BR \to BR$. We can then iterate to get a map $B^n R \times BR \to B^{n+1} R$. In general, a "multiplcation" map from $E_p \wedge E_q \to E_{p+q}$ induces a map

$$\tilde{E}^p(X) \times \tilde{E}^q(X) = [X, E_p] \times [X, E_q] \longrightarrow [X \wedge X, E_p \wedge E_q] \xrightarrow{\mu_*} [X \wedge X, E_{p+q}] \xrightarrow{\Delta^*} [X, E_{p+q}] = \tilde{E}^{p+q}(X).$$

Note that if the multiplication map μ is commutative up to homotopy, we get that the induced multiplication is commutative.

8.8. **Theorem.**

$$\tilde{H}_q(X; G) = \operatorname{colim}_q \left(\pi_q(X \wedge K(G,0)) \xrightarrow{\Sigma} \pi_{q+1}(X \wedge K(G,1)) \right)$$

Note that the above suspension map is given by the composition

$$\pi_q(X \wedge K(G,0)) = [S^1, X \wedge K(G,0)] \xrightarrow{\Sigma} [\Sigma S^q, \Sigma(X \wedge K(G,0))] = [S^{q+1}, X \wedge \Sigma K(G,0)] \longrightarrow [S^{q+1}, X \wedge K(G,1)]$$

9. What is the point of spectra? (Day 12)

What is the point of spectra? It attempts to unify the following

$$\begin{tikzcd}
\text{Stable information about spaces} \arrow[dr, rightsquigarrow] & & \\
& \text{spectra} \arrow[r, rightsquigarrow] & \text{algebraic structures} \\
\text{Generalized cohomology theories} \arrow[ur, rightsquigarrow] & & \begin{array}{l}\text{multiplicative structure on spectra}\\ \text{gives ring structure on cohomology}\end{array} \arrow[u]
\end{tikzcd}$$

Recall from Definition 8.2 that a spectrum is a sequence of spaces $E = \{E_n\}$ with maps $E_n \to \Omega E_{n+1}$; the suspension-loop adjunction gives maps $\Sigma E_n \to E_{n+1}$. This yields maps on homotopy groups

$$\pi_q(E_0) \longrightarrow \pi_{q+1}(\Sigma E_0) \longrightarrow \pi_{q+1}(E_1) \longrightarrow \pi_{q+2}(E_2) \longrightarrow \cdots$$

From this we can define homotopy groups for the spectrum E:

$$\pi_q(E) \coloneqq \operatorname{colim}_n \pi_{q+n}(E_n).$$

Additionally, recall that we did not need the full power of the fact that the map $E_n \to \Omega E_{n+1}$ is a homotopy equivalence—we just needed the fact that there is a map from $\Sigma E_n \to E_{n+1}$.

9.1. **Example.** Let $T_n = \Sigma^n X$ for some space X. Then $\Sigma T_n = T_{n+1}$ so we have maps $\Sigma T_n \to T_{n+1}$. If we take $\operatorname{colim}_n \pi_{q+n}(T_n)$ this is by definition the stable homotopy group $\pi_q^{\mathrm{st}}(X)$. We can view this as a special case of the stabilizing maps on the CW complexes X and Y:

$$[X, Y] \longrightarrow [\Sigma X, \Sigma Y] \longrightarrow [\Sigma^2 X, \Sigma^2 Y] \longrightarrow [\Sigma^3 X, \Sigma^3 Y] \longrightarrow \cdots,$$

Given this, we can take the colimit!

9.2. **Definition.** For CW complexes X and Y, we define $[X, Y]^{\mathrm{st}} = \operatorname{colim}_n [\Sigma^n X, \Sigma^n Y]$.

9.3. **Theorem.** *For CW complexes X and Y, the group $[X, Y]^{st}$ is abelian.*

Proof. We can represent $\alpha, \beta \in [X, Y]$ by $f, g \in \Sigma^k X \to \Sigma^k Y$ for $k \geq 2$. Then we can represent $\alpha + \beta$ by the composite

$$\begin{tikzcd}
\Sigma^k X \arrow[rr] \arrow[d, equal] & & \Sigma^k X \arrow[r, "f \vee g"] & \Sigma^k Y \\
S^k \wedge X \arrow[r, "\text{pinch!}"] & (S^k \vee S^k) \wedge X \arrow[d, equal] & & \\
& (S^k \wedge X) \vee (S^k \wedge X) & &
\end{tikzcd}$$

Details need to be checked and note that we need $k \geq 2$ for this construction to be abelian.

9.4. **Theorem** (half-joking). *We don't care about the definition of spectra.*

In particular, we don't care about the details of the definition, but we do care about the following. Things called *spectra* and homotopy classes of maps between spectra $[X,Y]^{\mathrm{Spec}}$ should satisfy

(1)
- A based space X gives a spectrum $\Sigma^\infty X$ coming from $\{\Sigma^n X\}$
- A spectrum E has an infinite loop space $\Omega^\infty E$ where $[\Sigma^\infty X, E]^{\mathrm{Spec}} \cong [X, \Omega^\infty E]$.

(2) If A is a finite CW complex and B a CW complex, then

$$[\Sigma^\infty A, \Sigma^\infty B]^{\mathrm{Spec}} = [A,B]^{\mathrm{st}}$$

(3) For a CW complex A and a spectrum X, we have spectra $X \wedge A$ and $F(A,X)$ which are adjoint:

$$[X \wedge A, Y]^{\mathrm{Spec}} \cong [X, F(A,Y)]^{\mathrm{Spec}}$$

So, spectra should be closely related to topological spaces and should encode their stable information.

(4) $[X,Y]^{\mathrm{Spec}}$ is an abelian group and it extends to a graded abelian group $[X,Y]^{\mathrm{Spec}}_\bullet$ such as

$$[X,Y]^{\mathrm{Spec}}_n := [X \wedge S^n, Y]^{\mathrm{Spec}}.$$

We also have a bilinear composition map $[Y,Z]^{\mathrm{Spec}}_\bullet \times [X,Y]^{\mathrm{Spec}}_\bullet \to [X,Z]^{\mathrm{Spec}}_\bullet$.

(5) There is a wedge product $X \vee Y$ that behaves like the direct sum of abelian groups.

(6) We have a construction $X \mapsto X \wedge S^1$ which is an invertible operation.

Properties 4 through 6 are stable/algebraic properties. Now we wish to discuss how spectra encode information about generalized cohomology theories.

(7) Every reduced cohomology theory is represented by a spectrum: $\tilde{E}^\bullet(A) \cong [\Sigma^\infty A, E]^{\mathrm{Spec}}_{-\bullet}$.

(8) Every spectrum defines a cohomology theory in this way

(9) A map $E \to F$ should give natural maps on cohomology $\tilde{E}^\bullet(A) \to \tilde{F}^\bullet(A)$.

These are amazing properties... but we want more. We want spectra to be perfect for doing any sort of algebra we could dream of. Let us really do algebra rather than just algebra up to homotopy.

(10) There exists a smash product of spectra $X \wedge Y$ such that

$$\Sigma^\infty A \wedge \Sigma^\infty B \cong \Sigma^\infty (A \wedge B)$$

for CW complexes A, B. We want this to behave like the tensor product of rings so we want $X \wedge Y = Y \wedge X$ and $\Sigma^\infty S^0 \wedge E \cong E$.

(11) For any spectra X, Y there exists a "function spectrum" $F(X,Y)$ such that $\pi_0 F(X,Y) = [X,Y]^{\mathrm{Spec}}$, where $\pi_0 A$ is defined to be $[\Sigma^\infty S^0, A]$. Additionally we should have

$$[X \wedge Y, Z] \cong [X, F(Y,Z)]^{\mathrm{Spec}}.$$

This is the best kind of product!!!!

(12) Each spectrum should define a homology theory via

$$\tilde{E}_q(X) = \pi_q(E \wedge X).$$

Tl;dr the definition of spectrum varies so we just say it must satisfy the above properties.

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